Antiderivatives and Riemann Sums

ANSWER KEY

Basic antiderivatives

1 Find the general antiderivative of each function:

a $f(x) = x^4 \quad = \frac{x^5}{5} + C$

b $f(x) = \frac{1}{x} \quad = \ln|x| + C$

c $f(x) = e^x \quad = e^x + C$

d $f(x) = \sec^2 x \quad = \tan x + C$

2 Find $\int (3x^2 - 4x + 5)\,dx$.
$= x^3 - 2x^2 + 5x + C$.

3 Find the particular antiderivative $F(x)$ of $f(x) = 6x^2 - 4$ satisfying $F(1) = 5$.
$F(x) = 2x^3 - 4x + C$. $F(1) = 2 - 4 + C = 5 \Rightarrow C = 7$. $F(x) = 2x^3 - 4x + 7$.

Riemann sums: left, right, and midpoint

4 Approximate $\int_0^4 x^2\,dx$ using a Left Riemann Sum with $n = 4$ equal subintervals.
$\Delta x = 1$; left endpoints $x = 0, 1, 2, 3$: $L_4 = 1[0^2 + 1^2 + 2^2 + 3^2] = 1[0 + 1 + 4 + 9] = 14$.

5 Approximate $\int_0^4 x^2\,dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.
$\Delta x = 1$; right endpoints $x = 1, 2, 3, 4$: $R_4 = 1[1 + 4 + 9 + 16] = 30$.

6 Approximate $\int_0^4 x^2\,dx$ using a Midpoint Riemann Sum with $n = 4$ equal subintervals.
$\Delta x = 1$; midpoints $x = 0.5, 1.5, 2.5, 3.5$: $M_4 = 1[0.25 + 2.25 + 6.25 + 12.25] = 21$. (True value: $\frac{64}{3} \approx 21.33$.)

The Trapezoidal Rule

7 Use the Trapezoidal Rule with $n = 4$ to approximate $\int_0^4 x^2\,dx$.
$T_4 = \frac{L_4 + R_4}{2} = \frac{14 + 30}{2} = 22$.

8 A table gives $f(0) = 2$, $f(2) = 5$, $f(4) = 9$, $f(6) = 8$. Use the Trapezoidal Rule with these unequal-width data (widths of 2 each) to approximate $\int_0^6 f(x)\,dx$.
With 3 equal subintervals of width 2:
$T = 2\left[\frac{f(0) + f(2)}{2} + \frac{f(2) + f(4)}{2} + \frac{f(4) + f(6)}{2}\right] = 2\left[\frac{7}{2} + \frac{14}{2} + \frac{17}{2}\right] = 7 + 14 + 17 = 38$.

Sums as limits: the definite integral

9 Explain how the definite integral $\int_a^b f(x)\,dx$ is defined as a limit of Riemann sums, and write the limit expression using sigma notation.

$\int_a^b f(x)\,dx = \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\Delta x$, where $\Delta x = \frac{b-a}{n}$ and x_i^* is any sample point in the i-th subinterval — as $n \to \infty$ (subintervals shrink to zero width), all Riemann sum types (left, right, midpoint) converge to the same value, the exact signed area under f.

10 Determine whether the Left Riemann Sum overestimates or underestimates $\int_0^4 x^2\,dx$, and explain why using the shape of the graph.

Since $f(x) = x^2$ is increasing on $[0, 4]$, the left endpoint of each subinterval gives the smallest function value on that subinterval, so the Left Riemann Sum UNDERESTIMATES the true area (confirmed: $L_4 = 14 < \frac{64}{3} \approx 21.33$).

Riemann sums from a table of values

11 A table gives $f(0) = 3$, $f(1) = 5$, $f(2) = 8$, $f(3) = 10$, $f(4) = 9$. Use a Right Riemann Sum with 4 subintervals of width 1 to approximate $\int_0^4 f(x)\,dx$.

$R_4 = 1[f(1) + f(2) + f(3) + f(4)] = 5 + 8 + 10 + 9 = 32.$

Antiderivatives of trigonometric and exponential functions

12 Find $\int (3\sin x + 2e^x)\,dx$.

$= -3\cos x + 2e^x + C.$

13 Find the particular antiderivative $F(x)$ of $f(x) = \sec^2 x$ satisfying $F(0) = 2$.

$F(x) = \tan x + C$. $F(0) = 0 + C = 2 \Rightarrow C = 2$. $F(x) = \tan x + 2$.